Mapping n Binary Inputs To 2^n Distinct Outputs

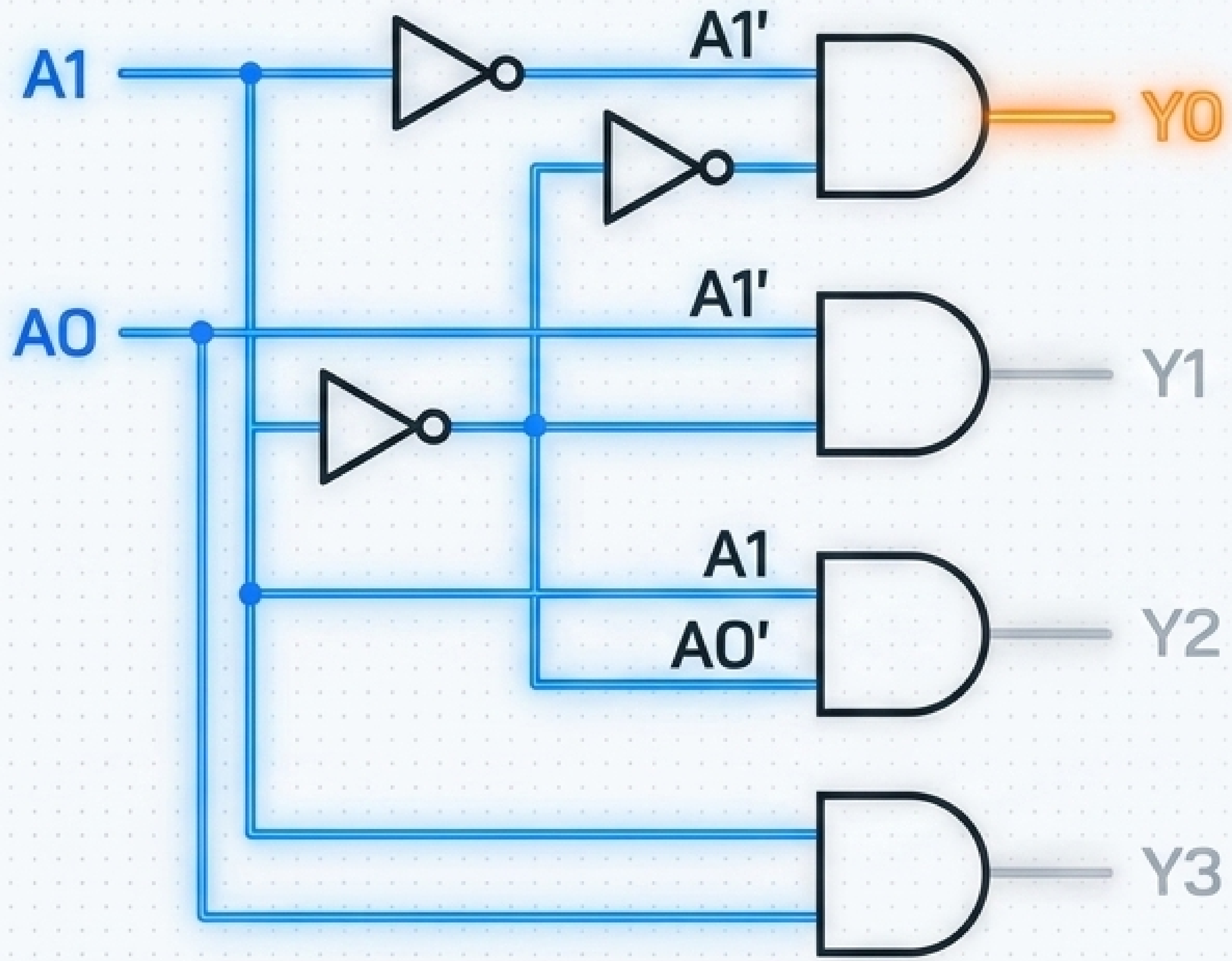
Combinational logic circuit converting n binary inputs into up to 2^n distinct outputs.

The golden rule: Exactly 1 output is active at 1 time.

If inputs change, the active output shifts instantly.

Translates compressed binary codes into 1 specific physical action.

Inside The Hardware Of The 2 To 4 Decoder



Built using exactly 4 AND gates and 2 NOT gates.	Inputs: A1, A0.	Outputs: Y0, Y1, Y2, Y3.

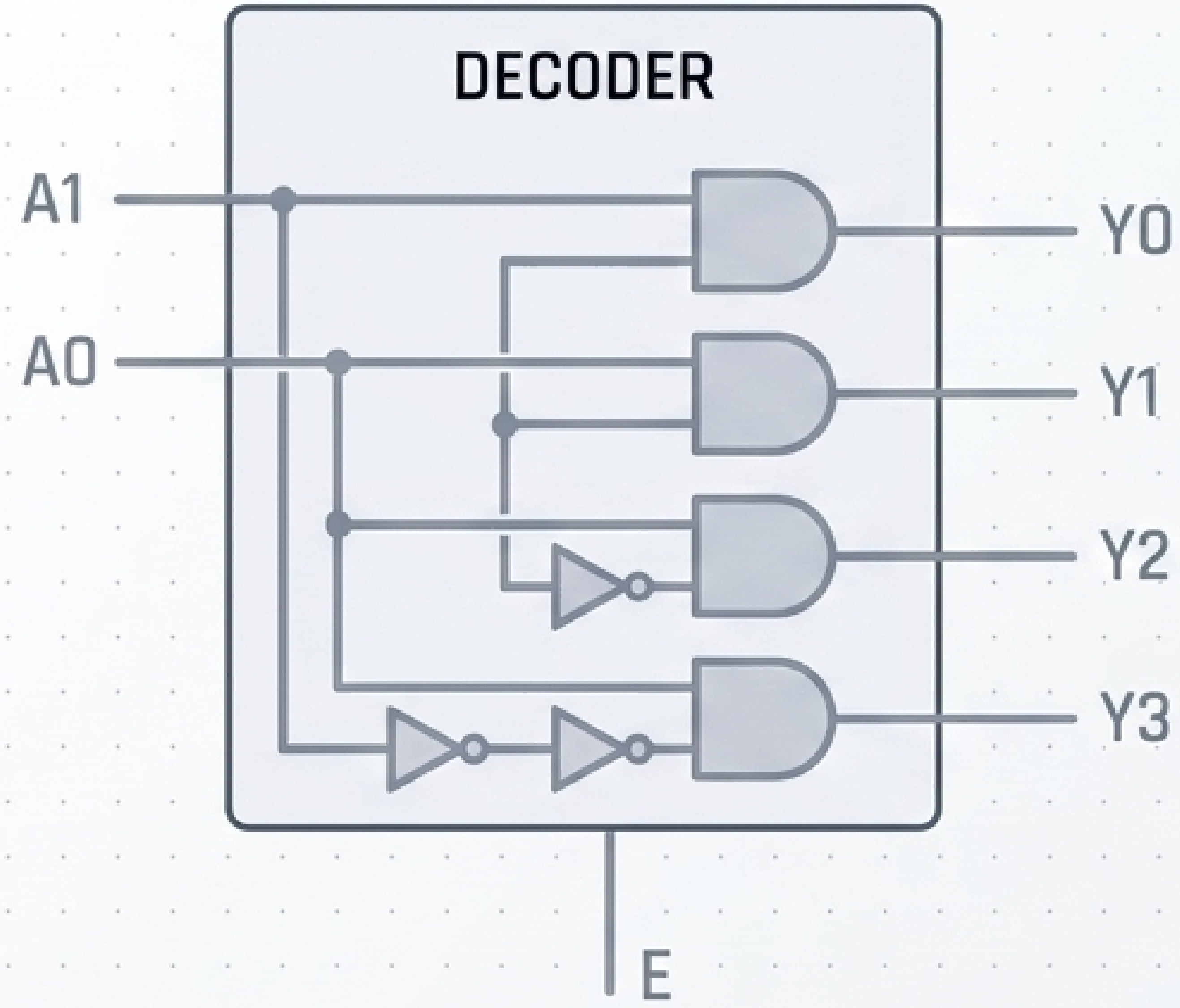
Y0 = A1' A0'

Y1 = A1' A0

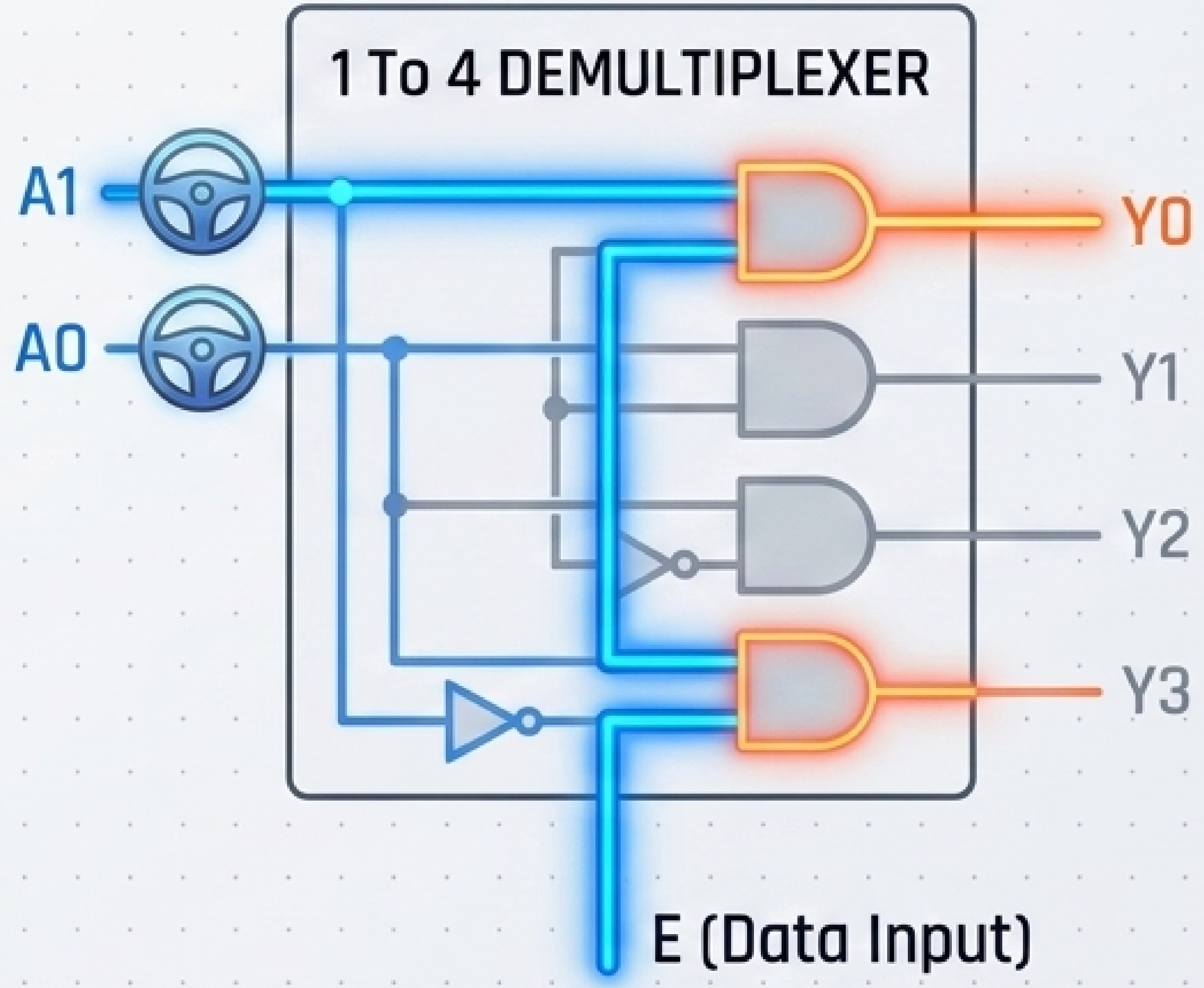
Y2 = A1 A0'

Y3 = A1 A0

Upgrading To 1 Demultiplexer With Enable Pins

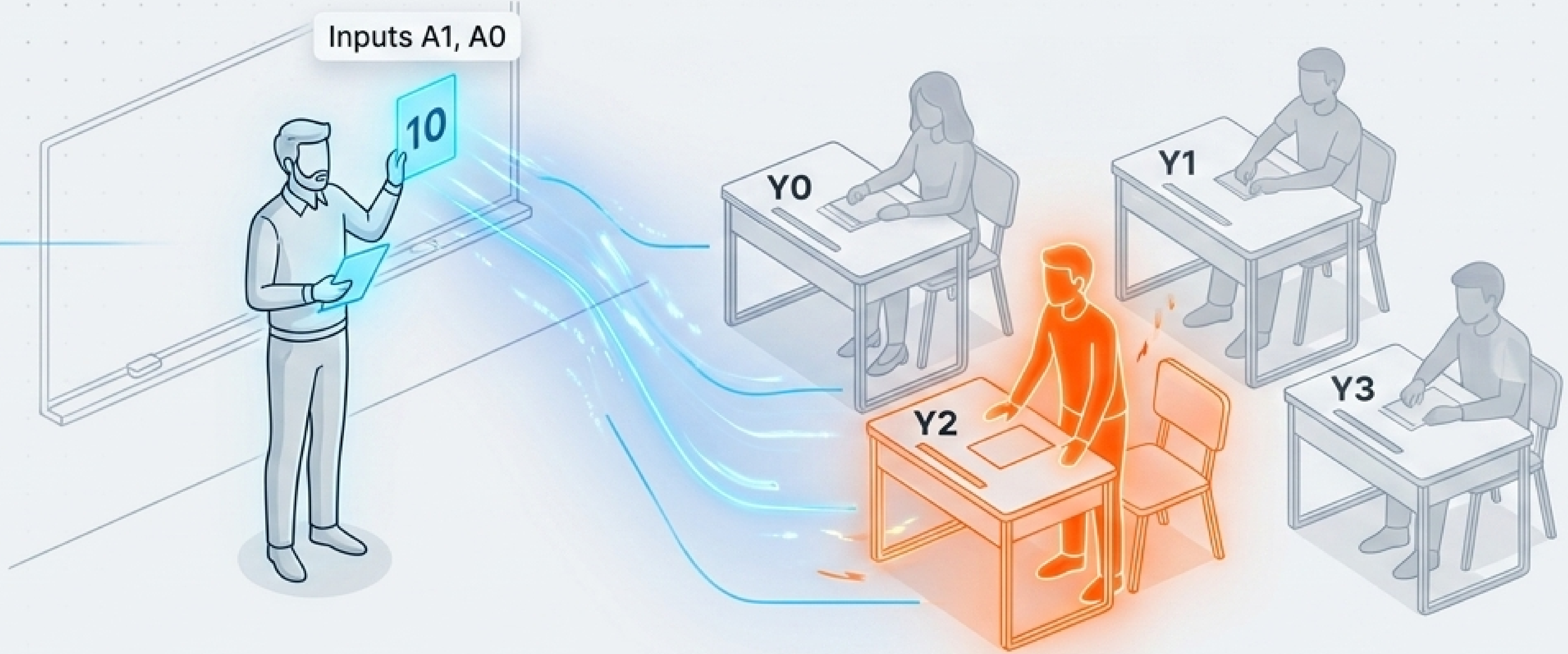


Adding 1 Enable (E) input controls the entire chip.
If E = 0: All outputs are 0.



If E = 1: Exactly 1 output mirrors the input state.
This identical circuit acts as 1 Demultiplexer.
E becomes the Data Input; A1 and A0 become Select lines.

Decoding The Math Using The Classroom Analogy



1.
Teacher (Inputs A1, A0) calls roll number 00, 01, 10, or 11.

2.
Student (Outputs Y0, Y1, Y2, Y3) waits for their exact ID.

3.
Teacher calls 10.

4.
Student Y2 stands up.

5.
Exactly 1 student responds.